

Generative AI (GenAI) - Quick Notes

1. What is AI?

- AI = Machines performing tasks that normally require human intelligence.
- Examples:
 - Face Recognition
 - Google Maps
 - Spam Detection

2. What is Generative AI?

- A type of AI that **creates new content**.
- Can generate:
 - Text
 - Images
 - Code
 - Audio
 - Video

Examples

- ChatGPT
- Gemini
- Claude
- Copilot

3. Traditional AI vs Generative AI

Traditional AI	Generative AI
Predicts	Creates
Classifies	Generates
Detects spam	Writes email
Recognizes image	Creates image

4. Large Language Model (LLM)

- LLM = Large Language Model
- Trained on huge amounts of text.
- Predicts the **next most likely word (token)**.

Examples:

- GPT
- Gemini
- Llama
- Claude

5. Token

A token is the smallest piece of text an LLM processes.

Example:

I love AI

Tokens:

I

love

AI

LLMs read and generate **tokens**, not complete sentences.

6. Prompt

A prompt is the instruction given to an AI.

Example:

Explain Java in simple words.

Better Prompt:

Explain Java in simple words with one real-life example.

Better prompt → Better output.

7. Prompt Engineering

Prompt Engineering is the process of writing **clear and effective prompts** to get better AI responses.

8. Context

Context is additional information given to AI.

Without Context:

Write an email.

With Context:

Write a professional leave email for two days due to fever.

9. Temperature

Controls creativity.

- Low (0–0.3) → More accurate
- Medium (0.5–0.7) → Balanced
- High (0.8–1.0) → More creative

10. Hallucination

When AI generates incorrect or made-up information confidently.

Always verify important information.

11. Embedding

An embedding converts text into numbers (vectors).

Purpose:

- Understand meaning
- Compare similarity
- Semantic search

12. Vector Database

Stores embeddings for fast similarity search.

Examples:

- ChromaDB
- Pinecone
- PGVector
- Milvus

13. RAG (Retrieval-Augmented Generation)

RAG = Retrieve + Generate

Steps:

1. User asks a question.
2. Search relevant documents.
3. Send documents + question to the LLM.
4. LLM generates a more accurate answer.

Benefit:

- More accurate
- Uses private/company data
- Reduces hallucinations

14. AI Agent

An AI Agent can:

- Think
- Plan
- Use tools
- Take actions
- Complete tasks automatically

Example:

Book a meeting -> Check calendar -> Find free time -> Send invitation

15. Common GenAI Applications

- Chatbots
- Code Generation
- Email Writing
- Document Summarization
- Translation
- Image Generation
- Content Creation
- Customer Support

- Question Answering

Prompt Engineering – Classroom Notes

What is a Prompt?

A **prompt** is the instruction or input given to an AI model to perform a task.

Examples:

Explain Java.

Write an email for leave.

Summarize this document.

The AI reads the prompt and generates a response.

What is Prompt Engineering?

Prompt Engineering is the process of designing clear, detailed, and structured prompts to get better responses from an AI model.

Definition:

Prompt Engineering is the art of writing effective prompts that help an AI produce accurate, relevant, and useful outputs.

Why is Prompt Engineering Important?

A vague prompt gives a vague answer.

A detailed prompt gives a better answer.

Example:

Poor Prompt

Explain Java.

Output:

- General explanation

Better Prompt

Explain Java for beginners in simple English with one real-life example.

Output:

- Beginner-friendly explanation
 - Simple language
 - Includes an example
-

Components of a Good Prompt

A good prompt usually contains:

- Task
- Context
- Role
- Constraints
- Output Format

Example:

You are a Java trainer.

Explain Java Collections for beginners.

Use simple English.

Give two real-life examples.

Limit the answer to 200 words.

Prompt Structure

Role

+

Task

+

Context

+

Constraints

+

Output Format

Example

Role:

You are an HR Manager.

Task:

Write an email.

Context:

Employee is applying for 2 days leave.

Constraint:

Professional tone.

Output:

Email.

Characteristics of a Good Prompt

A good prompt should be:

- Clear
- Specific
- Simple
- Complete
- Unambiguous

Avoid vague instructions.

Types of Prompting

1. Zero-Shot Prompting

The AI is given only the task.

Example:

Translate "Good Morning" into French.

No examples are provided.

2. One-Shot Prompting

Definition:

One example is provided before asking the AI to perform the task.

Example

English to Hindi

Hello to नमस्ते

Translate:

Good Morning

Expected Output

Good Morning -> सुप्रभात

3. Few-Shot Prompting

Definition:

Multiple examples are provided before asking the AI to perform the task.

Example

English -> Hindi

Hello -> नमस्ते

Thank You -> धन्यवाद

Good Night -> शुभ रात्रि

Translate:

Welcome

Expected Output

Welcome -> स्वागत है

Explanation:

The AI learns the translation pattern from the provided examples and applies the same pattern to translate the new word.

4. Role Prompting

Assign a role to the AI.

Example

You are an experienced Java Trainer.

Explain OOP concepts to first-year students.

The response is tailored to the assigned role.

5. Chain of Thought (CoT) Prompting

Ask the AI to solve a problem step by step.

Example

Solve this math problem step by step.

Useful for:

- Mathematics
- Logical reasoning
- Coding
- Problem solving

6. Structured Output Prompting

Request the output in a specific format.

Example

Return the response as JSON.

Example Output

```
{  
  "name": "John",  
  "age": 25  
}
```

Useful when integrating AI with applications.

Prompt Comparison

Weak Prompt

Write about AI.

Problems:

- Too broad
- No audience
- No length
- No format

Strong Prompt

Explain Artificial Intelligence to college students.

Use simple English.

Limit to 150 words.

Provide three real-life examples.

Best Practices

- Be specific.
- Give enough context.
- Mention the target audience.
- Mention the expected format.
- Mention the tone (formal, friendly, technical).
- Specify the output length if needed.
- Break complex tasks into smaller prompts.
- Refine prompts based on the AI's response.

Common Mistakes

- Asking vague questions.
- Giving incomplete information.
- Combining too many tasks in one prompt.
- Not specifying the output format.

- Assuming the AI knows hidden context.

Real-Life Examples

Email Writing

Poor Prompt : Write an email.

Better Prompt : Write a professional leave email requesting two days leave due to fever.

Coding

Poor Prompt : Write Java code.

Better Prompt : Write a Java 21 program to reverse a string using StringBuilder with comments.

Summarization

Poor Prompt : Summarize this.

Better Prompt : Summarize this article in five bullet points suitable for beginners.

SQL

Poor Prompt : Write SQL query.

Better Prompt : Using the HR schema, write an SQL query to display department-wise employee count sorted by department_id.

Benefits of Prompt Engineering

- Better quality responses
- Higher accuracy
- Saves time
- Reduces repeated prompting
- Easier automation
- Produces consistent outputs

RAG (Retrieval-Augmented Generation) – Short Notes

What is RAG?

RAG (Retrieval-Augmented Generation) is a technique that improves LLM responses by **retrieving relevant information from external data before generating an answer**.

Formula:

Retrieve -> Augment -> Generate

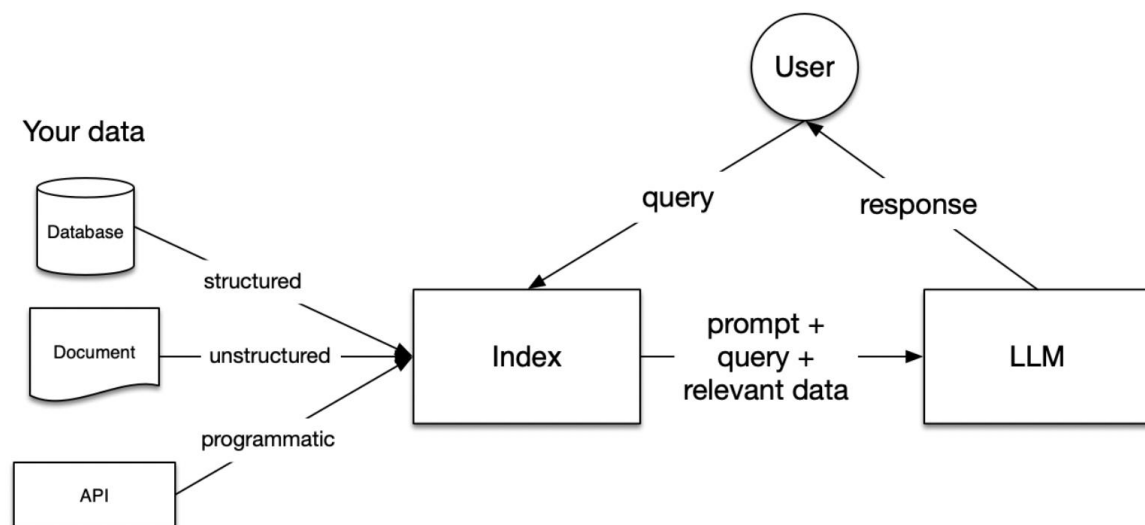
Why RAG?

LLMs:

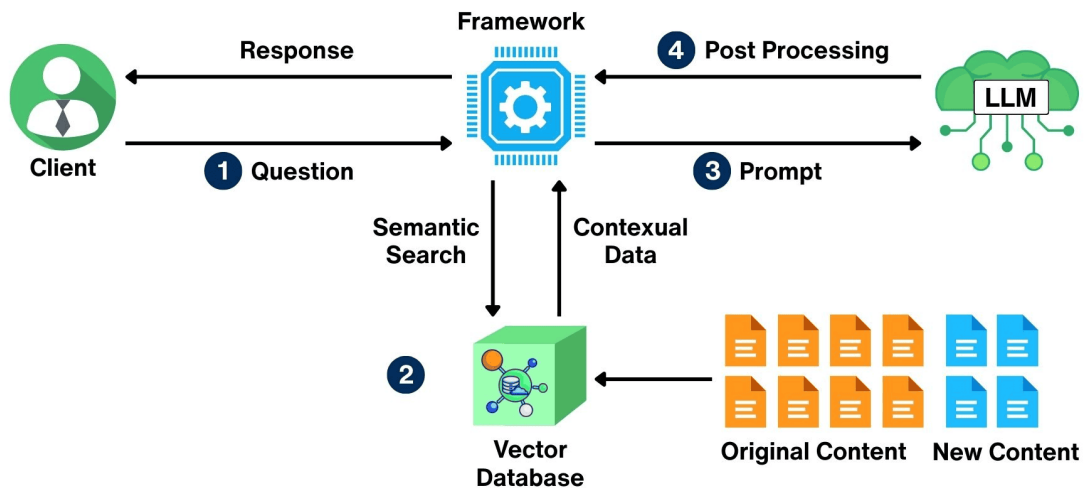
- Have limited knowledge after training.
- Cannot know your company's private data.
- May produce hallucinations (incorrect answers).

RAG solves these problems by providing relevant documents to the LLM.

RAG Workflow



RAG ARCHITECTURE MODEL



User Question



Convert Question to Embedding



Search Vector Database



Retrieve Similar Documents



Send Question + Documents to LLM



Generate Accurate Answer

Main Components

- **User Query** – User's question.
 - **Embedding Model** – Converts text into vectors.
 - **Vector Database** – Stores document embeddings.
 - **Retriever** – Finds the most relevant documents.
 - **LLM** – Generates the final answer.
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Advantages

- More accurate answers.
 - Reduces hallucinations.
 - Uses latest information.
 - Works with private/company documents.
 - No need to retrain the LLM.
-

Limitations

- Depends on document quality.
 - Poor retrieval leads to poor answers.
 - Slightly slower than using only an LLM.
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Real-Life Example

Question:

What is our company's leave policy?

Without RAG:

LLM guesses a generic leave policy.

With RAG:

Searches the company's HR policy documents -> Retrieves the correct policy ->

LLM answers based on the retrieved document.

Common Use Cases

- Company Chatbots
- HR Policy Assistant
- Customer Support
- Legal Documents
- Medical Knowledge Base
- Internal Documentation Search